

A large, stylized 'D' graphic composed of three overlapping semi-circular shapes in a lighter shade of orange, positioned in the upper left background.

Reporting Service API documentation

Technical guide to using the Reporting API

API version 2.0 | March 2025

Document version history

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What is the Reporting Service API?

Dayshape comes with rich reporting functionality that users can utilize for building user-friendly, highly configurable reports.

Dayshape also exposes this API so that customers can use it to build custom integrations. You can think of it as a wrapper over the application data layer.

The reporting functionality in Dayshape is driven by the dedicated Reporting Service, which is deployed alongside the Dayshape core application. Historically, reporting was embedded in the core application, but this approach led to performance bottlenecks and prevented scalability.

Within this document, any references to “legacy reporting API” are referring to the API within the core application. “Reporting Service API” relates to the new versioned APIs available within the reporting service.

This document provides information to help engineers interact with the Reporting Service API.

Authorization with Reporting Service API

Acquiring auth tokens

Before you can acquire a token, you will need to setup an API user within Dayshape. You should consider the reports you intend to use with this user and ensure that they are authorized to use them.

If any requests return a **403** (Forbidden) error, it is an indication that your API user does not have the necessary permissions to perform your request. Please reach out to the Dayshape Support or Professional Services team if you need any assistance with this,

Any tokens acquired will have an expiration date. This is configurable within the issuer of your tokens, but as a best practice you should anticipate that tokens will expire after 1 hour. If any API request returns a **401** (Unauthorized) response you should get a new token and retry the request.

Acquire self-issued tokens

Both Dayshape and the Reporting Service are capable of issuing their own tokens. Simply pass valid user credentials in a request to the Dayshape **POST api/token** route and you will be returned a valid token to use for protected API endpoints:

```
curl -X POST --location 'https://dayshape.app/reporting/token' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "apiuser@dayshape.com",
    "password": "#EBHCvnHcytLt9=v05z"
  }'
```

The response will include the token you requested:

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMjM0NTY3ODkwIiwibmFtZSI6IkpvaG4gRG9lIiwiaWF0IjoxNTE2MjM5MDIyfQ.SflKxwRJSMeKKF2QT4fwpMeJf36POk6yJV_adQssw5c
```

Authorization

The Dayshape Reporting Service API supports the HTTP Bearer Authentication scheme.

Simply supply a token you have already acquired and include it via the Authorization header, making sure to specify the bearer authentication scheme:

```
curl -X POST --location 'https://dayshape.app/reporting/v2'
-H 'Authorization: Bearer
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMjM0NTY3ODkwIiwibmFtZSI6IkpvaG4gRG9lIiwiaWF0IjoxNTE2MjM5MDIyfQ.SflKxwRJSMeKKF2QT4fwpMeJf36POk6yJV_adQssw5c'
-H 'Content-Type: application/json'
-d '{
  "reportId": "taskListing",
  "timeZone": "Europe/London",
  "from": "2021-05-02T00:00:00.000Z",
  "to": "2021-11-02T23:59:59.999Z",
  "filters": [],
  "dimensions": [
    {"dimensionId": "TaskId", "sorted": "ascending"},
    {"dimensionId": "TaskStart"},
    {"dimensionId": "TaskEnd"}
  ]
}'
```

API versioning

The Reporting Service API makes use of API versioning so that Dayshape can manage multiple versions of any RESTful APIs in a single service. This enables Dayshape to make changes to the Reporting Service without breaking any customer integrations. Please note that the API Version is not the same as the version of the Dayshape application, which is of the form vxx.x.x.x.

The service uses a *major* and *minor* version scheme, which must be supplied to versioned APIs using URL path versioning:

`https://dayshape.app/reporting/v{major}.{minor}`

When the minor version is 0, the minor number can be omitted from the URL. For example, the API will treat requests to either of the following URLs as being for API version v2.0:

`https://dayshape.app/reporting/v2.0`

`https://dayshape.app/reporting/v2`



When a version is **deprecated**, it means that Dayshape no longer maintains that functionality and intends to eventually sunset the version. When a version is **sunsetted**, it is no longer available for use and the endpoint no longer exists.

Dayshape strongly recommends that any API consumers upgrade to the latest supported version as soon as possible. Dayshape will provide consumers with advance warning before an API version is to be deprecated. These reminders will continue to be communicated until all consumers have migrated from the deprecated version or until the version has been sunsetted.

Version discovery

Versioned Reporting Service APIs return two response headers to help with version discovery:

1. **api-supported-versions** – comma-separated list of versions supported. These are being proactively maintained by Dayshape.
2. **api-deprecated-versions** – comma-separated list of versions that have been deprecated (see note below).

Example response:

```
HTTP/2 200 OK
...
Api-Supported-Versions: 1.0, 2.0, 3.0
Api-Deprecated-Versions: 0.9
```

The same list of versions will be returned whenever a request is made to a versioned endpoint, regardless of whether the version used is deprecated or supported.

Version sunsetting

The Reporting Service API supports [RFC 8594](#). This allows an API version to indicate when it will be removed for good via the **Sunset** header.

```
HTTP/2 200 OK
...
Api-Supported-Versions: 1.0, 2.0, 3.0
Api-Deprecated-Versions: 0.9
Sunset: Tue, 01 Jan 2030 00:00:00 GMT
```

The sunset header does not apply to all API versions; it will only apply to the API version that was requested.

Versions supported

At the time of writing, the following versions are currently supported.

Version	Description
v2.0	The v2.0 API provides richer control to the consumer, allowing any callers to specify the output format, locale, time zone etc. Both request and response payloads have been redesigned in this new version. Additionally, the names of <i>filters</i> and <i>dimensions</i> accepted by the run report endpoints have been changed so they better describe their behavior.
v1.0	This version has been deprecated and will sunset in July 2025. Do not use this version. This should not be used for any new integrations, and is only designed as an adapter layer to allow customers to continue to use their existing legacy payloads while they migrate their integrations to v2.0. Now that the legacy service is deprecated, use of this version is no longer recommended , and we advise all customers to migrate to v2.0 as soon as possible .



Ideally customers should migrate to new API versions as soon as possible, but we recognize this is not always possible. If the version you are using has been deprecated, it is a sign that you should migrate before the sunset date.

Migrating between versions

Upgrading from a deprecated API version

Each API version can accept and return different payloads. This means payload properties can be added, renamed or even removed entirely. As such, you need to make sure you can build and handle a version's payloads before migrating to that version.

There are several actions required by consumers to migrate:

1. Update your request payloads – this can be done either manually or by rebuilding the payload via the Dayshape UI (see below).
2. Update request path to use the new version.
3. Update your response handling – this must be done manually by reviewing the API specification for the API version you are migrating to.

Migrating to the Reporting Service v2.0



The Reporting Service v2.0 is now ready for use. Please use the Reporting Service v2.0 for all new integrations from this point onwards.

Payload differences

There are significant differences in the payloads between the legacy reporting API and the Reporting Service API. Structurally, the JSON is similar, however, there are significant changes to the naming of filters and dimensions which mean the old payloads are no longer compatible.

Using Dayshape to obtain a payload

Given the differences in payloads, there is no automated migration path to updating your payloads in the Reporting Service API.

However, you can use the Dayshape application to generate a payload which you can then pass via the API. This is outlined in the section [Manually Building Payloads](#) below.

Example requests

Example requests are displayed below. Please note, these are not real URLs.

Request URL	V2.0 URL
Get a Token	POST https://dayshape.app/reporting/token
Run a report	POST http://dayshape.app/reporting/v2.0

For full details, or if a request is failing, please see the provided API specification, https://<your-instance-name>.dayshape.app/reporting/swagger/v2/dayshape_reporting_api_spec_v2.json

Working with the Reporting Service API

Key concepts

All report queries, regardless of API version, take 3 key sets of properties:

1. **ReportId** – this is the type of the report you want to run. Report names in v1 and v2 APIs differ, so make sure to select the correct identifier.
2. **Filters** – These are used to filter to just the data you are interested in. These are applied to the initial SQL queries that pull data out of the database.
3. **Dimensions** – these are the *columns* that you want to be returned in your report results, and these values will be formatted based on the supplied formatting options.

Reports, filters, and dimensions can be utilized by consumers via unique identifiers.

Dayshape API spec

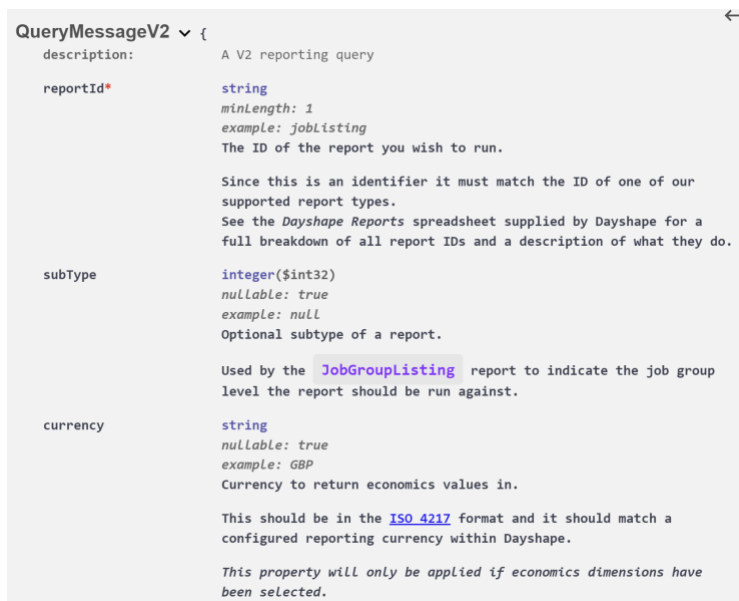
To make interacting with the Dayshape Reporting Service API as straightforward as possible, Dayshape provides a Swagger page to view the specification of the API. This is available at <https://<your-instance-name>.dayshape.app/reporting/swagger>.

From the Swagger page, you can find the following OpenAPI v3 specifications that describe the routes consumers are permitted to interact with:

https://<your-instance-name>.dayshape.app/reporting/swagger/v2/dayshape_reporting_api_spec_v2.json

Customers can also load the OpenAPI Endpoint directly into Postman to generate a collection to send example requests to the Reporting Service API.

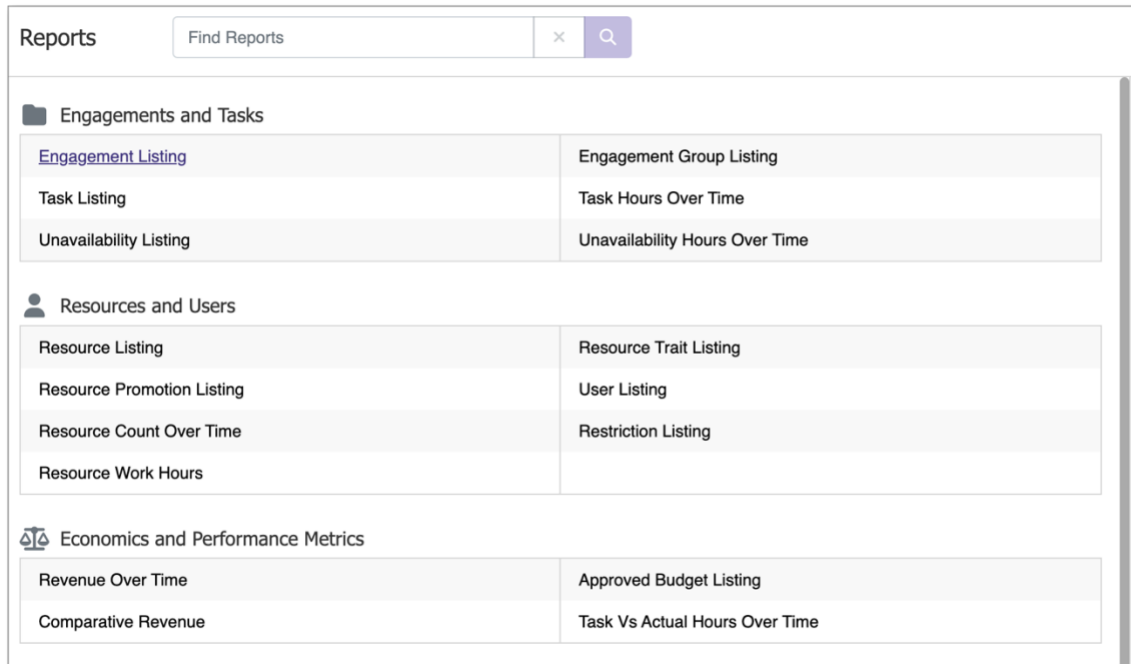
The specifications include a full breakdown of all payloads that are accepted and produced, so consumers can use these to adjust any payloads that have been built via the Dayshape UI. For example, the following screenshot shows you how the payload definition would appear in Swagger:



Generating payloads

Given the volume of identifiers and the huge number of permutations accepted by the Reporting Service API, we strongly recommend that all consumers build request payloads via the Dayshape UI.

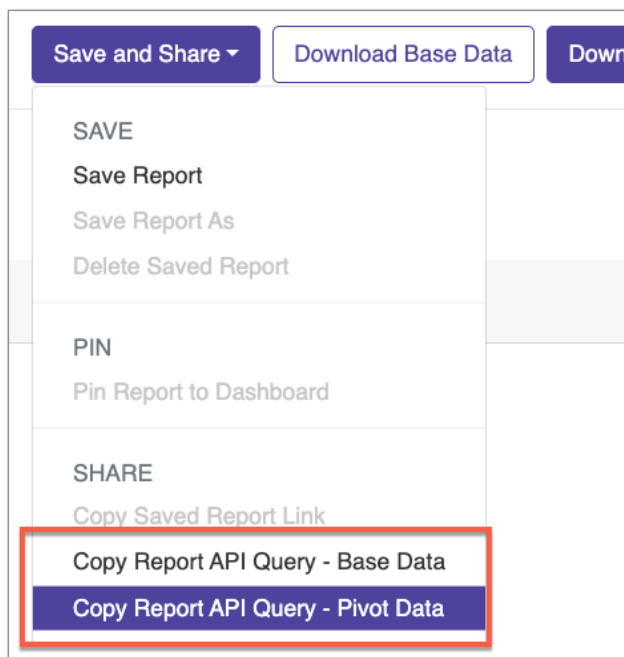
To make use of this functionality, navigate to the report selection page and select the type of report you want to build a query for:



All reports will build v2.0 payloads from the release of Dayshape 23.8.

On the report builder page, select all the date options, filters, and columns/pivot fields that you require for the report query you want to execute. Once you're ready, click the **Save and Share** drop-down and select **Copy Report API Query**.

If you are using a pivot report, you will be given two options: **Copy Report API Query – Base Data** and **Copy Report API Query – Pivot Data**.



The payload will be copied to your clipboard, ready to be pasted into a HTTP request within your HTTP client of choice. You can also modify the payload at this stage to contain any dynamic information (e.g., apply parameterized inputs).



This functionality will only be available to users with the `Reporting_SaveQueryDefinitionToClipboard` permission. If you can't see this functionality, change your permissions to reflect the above, or please contact Dayshape Support.

Manually building payloads

You can also manually build your payloads. To aid consumers, Dayshape have built the spreadsheet **Dayshape Reporting Service Detail.xlsx** which covers the reports that Dayshape supports. Consumers should refer to this when modifying existing payloads or if building payloads from scratch. This spreadsheet is updated whenever changes to filters and dimensions are released and is versioned with Dayshape's core application. Please ensure your spreadsheet version corresponds with your version of Dayshape.

The landing sheet of **Dayshape Reporting Service Reports.xlsx** provides a breakdown of all reports supported by the reporting service, along with links to specific sheets describing each report's supported functionality.


Dayshape

Dayshape Reporting Service

This spreadsheet describes all reports available in Dayshape, including further information on the permissions needed, data visibility, and the columns and filters available for each report.

 This spreadsheet is written using Dayshape standard terminology which reflects the default UK language pack. The Dayshape application you use will display your own glossarised terms, which may differ slightly from the terminology used throughout this document.

 The term "Columns" is generally used in this spreadsheet to refer to reporting dimensions that can be selected for building a Listing report. "Pivot Fields" is generally used in this spreadsheet to refer to reporting dimensions that can be pivoted on when building a pivot report (includes both pivot rows and pivot columns). These are displayed in the Dayshape UI (subject to glossarisation). The term "Dimensions" (where used in this spreadsheet) refers to the static and unglossarised reporting dimension. For example, in the Engagement Group Listing report using the default UK language pack, the column 'Recovery Rate %' represents the reporting dimension 'JobGroupStatsRecoveryRate'. PLEASE ALSO NOTE: In listing type reports, the selectable entities "Columns" are used as "Rows" on which to pivot.

 The information in this spreadsheet is accurate based on columns/dimensions and filters that are available in the Reporting Service API, but not all of those entities are currently available when building a report from the UI. We are currently addressing these gaps on a release-by-release basis.

Engagements and Bookings			Reports which provide information on Engagements, Engagement Groups, and Bookings. Particularly helpful for users tracking activity, Economics, or progress across a large portfolio of work.
Report Name	Report Type	Link	
- Engagement Listing	Listing	Link	
- Engagement Group Listing	Listing	Link	
- Booking Listing	Listing	Link	

Each specific report sheet describes all columns/dimensions and filters supported by that report type. The following spreadsheet columns can be used to get identifiers you can use in your payloads:

1. **Dimension Id** – Reporting Service Dimension identifiers for use with v2.0 endpoints.
2. **Filter Id** – Reporting Service Filter identifiers for use with v2.0 endpoints.
3. **Filter Input** – Reporting Service Filter parameters that a given filter will accept.

There are also examples of the legacy dimensions and filters used in the v1.0 endpoints. This can help you confirm where changes have been made to identifiers between v1.0 and v2.0.

Report results formatting

The Reporting Service APIs accept formatting options within their payloads for controlling how the results of a report should be formatted. This will be useful in enabling customers to reduce the amount of formatting needed in any custom integrations.

Users can request differences to the following formatting options in v2 payloads:

- Timezone
- Locale
- Date Format
- Time Format
- Number Formats
 - By default, numbers are unformatted and unrounded (as of Dayshape v25.2.0.0)
 - You can choose whether numbers are formatted with the variable `formatNumbers`
 - If **false**:
 - Numbers return unformatted and unrounded
 - Percentages return unformatted and unrounded. Will be expressed in decimal, e.g. '0.9'
 - Economic values return unformatted and unrounded
 - If **true**:
 - Numbers return formatted and rounded to 2 decimal places and, if specified, will be localized (e.g. using commas instead of periods for the decimal point)
 - Percentages return formatted (e.g. 90%) and rounded to 2 decimal places and, if specified, will be localized (e.g. placing the percentage sign before the value)
 - Economic values will be rounded to a whole number and, if specified, will be localized (e.g. using periods when representing thousands)
 - We recommend customers have `formatNumbers` set to **false** and provide this by default when API queries are copied using **Copy Report API Query** in the UI
- Null Formats

Full details on formatting controls can be found in the supplied OpenAPI specifications in the `QueryMessageV2` Schema.

Data freshness

The reporting service uses cached data to speed up report query execution times. The cached data is refreshed every two minutes, meaning that when a report query is executed, the results returned might not be completely up to date.

To convey how fresh the data in the report is, the custom response header **Dayshape-Oldest-Cache-Age** is returned:

```
HTTP/2 200 OK
...
Dayshape-Oldest-Cache-Age: Fri, 21 Jul 2023 14:07:01 GMT
```

Timeouts

The nature of the Reporting Service means that some requests for significant ranges of data mean that reports sometimes do not return in a timely manner. We have set a timeout of **15 minutes** maximum per request. For most customers, this will be ample time, but if you are regularly finding that your requests time out, please raise a Freshdesk ticket.

Rate limiting

The Reporting API has rate limiting built in for some reports. For the reports below, a single user cannot run more than one of the below reports at a time, otherwise you will receive an HTTP 429 Response.

These reports are:

- Availability
- Availability Planning
- Booking Audit Trail
- Booking Hours Over Time
- Booking Listing
- Booking Vs Actual Hours Over Time
- Comparative Revenue
- Clash Hours Over Time
- Engagement Audit Trail
- Engagement Group Audit Trail
- Resource Audit Trail
- Resource Work Hours
- Revenue Over Time
- Settings Audit Trail
- Unavailability Hours Over Time
- Unavailability Audit Trail
- User Audit Trail

For example, if you run a Booking Hours Over Time, then attempt to run a Resource Work Hours report before Booking Hours Over Time has completed, you will receive an HTTP 429 Response. This only applies to individuals – another user running another of these reports at the same time as you will not have an impact.

Best practices

Filters for pagination

The Reporting API does **not** support pagination. As an alternative, we encourage users to apply filters (e.g. rolling date filters) to requests if needed. This is particularly recommended if you have any downstream limitations on ingesting large API responses.

For example, rather than pulling 2 years of data in a single query, consider running 2 report queries, where each report query is building a report for a single year.

So, rather than sending this single request:

```
curl -X POST --location 'https://dayshape.app/reporting/v2'
-H 'Authorization: Bearer
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMjM0NTY3ODkwIiwibmFtZSI6IkpvaG4gRG9lIiwiaWF0IjoxNTE2MjM5MDIyfQ.SflKxwRJSMeKKF2QT4fwpMeJf36POk6yJV_adQssw5c'
-H 'Content-Type: application/json'
-d '{
  "reportId": "taskListing",
  "timeZone": "Europe/London",
  "from": "2020-01-01T00:00:00.000Z",
  "to": "2021-12-31T23:59:59.999Z",
  "filters": [],
  "dimensions": [
    {"dimensionId": "TaskId", "sorted": "ascending"},
    {"dimensionId": "TaskStart"},
    {"dimensionId": "TaskEnd"}
  ]
}'
```

You could make two requests, one for each year:

1. The first request would run from 1st January 2020 to 31st December 2020:

```
"from": "2020-01-01T00:00:00.000Z",
"to": "2020-12-31T23:59:59.999Z",
```

2. The second request would run from 1st January 2021 to 31st December 2021:

```
"from": "2021-01-01T00:00:00.000Z",
"to": "2021-12-31T23:59:59.999Z",
```

Use as few dimensions as possible

The more data you're pulling, the slower your report query will execute. Only use the dimensions that your desired use case requires.

Filter to what you care about

Generally speaking, the more data you return, the slower a query will execute. Only filter the data you require. Be aware that too many filters in a query can impact performance, so try to stick to one or two filters in any given query.

Expect data to change

Consider how data will change when you build your report queries. The goal is to future-proof your custom reports so that you don't need to spend too long maintaining them. For example, rather than reporting on resources by their email address, report on resources by grade, as resources can change over time. Or if you are filtering by units and you plan to change unit names, then you may want to reconsider your query.

Use identifiers in response handlers

Request response handlers should be built to use *identifiers*. For example, v2.0 endpoints can return dimensions with their localized names, but the *identifier* of the dimensions should access specific dimensions within your handler logic. Identifiers are guaranteed to be static but other fields are not.